How did Ivory Coast win AFCON 2023 despite sacking their manager during the Tournament, and was it deserved?

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1 My Resources

Live dashboard: https://nahal12345.github.io/Sports-Data-and-Analytics-Assignment-/

Repository: https://github.com/Nahal12345/Sports-Data-and-Analytics-Assignment-

2 Introduction

AFCON 2023 was hosted in Côte d'Ivoire (Ivory Coast), and it saw the hosts lift the title. This victory was remarkable not just because it was a home tournament, but the team were on the brink of elimination during the group stage. After their win in the first game, Ivory Coast followed that by losing two consecutive games to Nigeria and Equatorial Guinea which saw them finish in 3rd place, narrowly qualifying due to them being the 4th best 3rd place finishers. Due to this failure in the group stage, the Ivorian Football Federation took the decision to sack the head coach Jean-Louis Gasset and replace him with Emerse Faé, who was the Ivory Coast under 23 coach.

Following this there was an unexpected revival, with Ivory Coast which led to AFCON victory. Faé's side beat Senegal in the Round of 16, Mali in the Quarter Final, Democratic Republic of Congo in the Semi Final, and then defeated Nigeria in the final, which was one of the teams that they had previously lost to in the group stage. They did not just beat them, they beat them convincingly as they dominated the final. The transformation between the two managers was visible, but the question that this report seeks to answer is whether the data supports the outcome: How did Ivory Coast win AFCON 2023 despite sacking their manager during the Tournament, and does the data back them up?

The question is interesting because football tournaments can be won due to pure luck and individual brilliance whilst also posting poor underlying numbers. In contrast, a dominant team can lose to a single moment . Expected Goals (xG), defensive action metrics, attacking metrics and shot quality data offers a way to cut through the narrative and allows us to analyse what exactly is happening with the players on the pitch.

The dashboard and the accompanying analysis is structured around the comparison between the group stage performance under Jean-Louis Gasset and the Knockout Stage performance under Emerse Faé, which individual players drove the improvement, and underlying data suggesting whether the title was deserved. The data that has been used throughout this report was sourced from StatsBomb's open data repository, which has provided event-level data for the AFCON data. The dashboard is hosted publicly on GitHub Pages and built using R, 'ggplot' and 'Plotly', with the full code available in the repository that has been linked.

3 Data Collection and Preparation

3.1 Data Source

All the data that has been used in the analysis was obtained from StatsBomb's open data repository, which was computed into R via the 'statsbombr' package (StatsBomb, 2024). This open data provided event level match data for AFCON, which allowed for extensive game by game analysis to be conducted. This was downloaded by: devtools::install.github("statsbomb/Stats

The StatsBomb event data model captures every on the ball action in a match as its own event, each with a time stamp, location (expressed as an x,y coordinate on a 120 x 80 pitch), the player and the possession team involved, and many more action specific events. The three primary functions that were used to collect the data were ‘FreeCompetitions()‘, ‘FreeMatches()‘ and ‘get.matchFree()‘. ‘FreeCompetitions()‘ was used to retrieve the full list of available competitions and identified AFCON2023 (StatsBomb, 2024). To filter for this, the competition id was filtered using 1267. This allowed for the match data to be downloaded, which is where ‘FreeMatches()‘ was used. In order to pull event level data from the matches, ‘get.matchFree()‘ was used to return a database of every recorded event, such as shots, passes, dribbles and duels etc.

3.2 Dataset structure and coverage

The final merged dataset consisted of all 52 AFCON 2023 matches. The event level data frame consisted of 162,910 rows across all matches, with each row representing a single recorded event. The key variables used in this research were:

Variable	Type	Description
‘match_id‘	Integer	Unique match identifier
‘competition_stage.name‘	Character	Round of competition
‘team.name‘	Character	Team Name
‘type.name‘	Character	Event type
‘shot.statsbomb_xg‘	Double	Expected goals value
‘shot.outcome_name‘	Character	Shot result
‘player.name‘	Character	Full player name
‘location‘	List	x,y coordinates on the pitch
‘minute/second‘	Integer	Event time stamp
‘possession_team.name‘	Character	Team in possession
‘duel.type.name‘	Character	Type of duel occuring
‘pass.goal_assist‘	Logical	Whether the pass led to a goal

For Ivory Coast specifically, the ‘ivorycoast_matches‘ subset identified the seven matches that they played in. The events from these matches were filtered from all_events using the ‘match_id‘ to analyse specific matches. This allowed for specific event data to be pulled from their matches, and using ‘possession_team.name == “Côte d’Ivoire”‘ allowed for a more focused view on the team.

One important structural decision that was required to be made was the exclusion of “period ==5”, which corresponds to a penalty shootout in Statsbomb databases. This was done as the penalty shootout goals, shots and xG were being included into the overall shot and goal analysis. Including these events would have altered the results, as penalty shootouts occur outside of normal time, therefore they do not represent in game performance in the same way that the regular event data does. Penalties carry a very high xG value, which would mean that when analysing the cumulative xG for Ivory Coast throughout the tournament, the penalty shootout vs Senegal would have interfered with the data collection.

3.3 Cleaning and Preparation

Before any of the metrics were computed, data cleaning and preparation steps were applied to ensure that the raw data was suitable for the analysis that was being prepared.

Spatial coordinates in the StatsBomb dataset are stored as list columns rather than standard numeric vectors, with each of the event data's location being represented by an x and y value. For the simple extractions, 'sapply'was used to pull the first and second elements of each location list. For the larger extractions where columns could contain empty entries, 'map_dbl' was used to substitute NA where the coordinates were missing. This ensured that any missing coordinate values did not cause any incorrect results within the analysis.

Shot outcomes were recoded from StatsBomb's categories into four different outcomes to be used across the shot maps. To keep the shot maps clean and readable, outcomes such as 'Saved to Post' were deliberately combined into 'Saved' rather than keeping it as it's own outcome. The four outcomes used were 'Saved', 'Blocked', 'Off Target' and 'Goal'. Shots were also filtered to exclude 'period 5' , which represents the penalty shootout. Including shots from penalty shootouts would have distorted the shot location data, as it would have inflated shot totals without affecting in game events.

A binary manager variable was created directly from 'competition_stage.name', which assigned the Group Stage matches to Jean-Louis Gasset and the Knockout Stage to Emerse Faé. This allowed for a direct comparison between the managers across each metric in the dashboard without having to go game by game. It meant that every match was assigned to the correct manager in the same way across every single section of the analysis.

3.4 Limitations of the Data

There are certain limitations of the dataset which affected the research. Firstly, the Statsbomb open data does not include tracking data (player positions in between the events), which limits the amount of off the ball metrics. This includes the pressing intensity and the overall defensive structure. Linking to this, the xG model used will not take underlying metrics such as pressure on the attacker, or how close the defender is, meaning the calculated xG values may not fully reflect the true quality and difficulty of the chance. Some of the results may be overvalued and some may be undervalued compared to the actual context of the shot.

The data does not take into account game state, meaning that events occurring when a team is winning, drawing or losing are all treated equally. This is a relevant limitation to the data because different circumstances in games cause different outcomes. For example, teams who are losing will take a lot more risks, which could lead to them taking more shots out of desperation, take more risks in passing and press higher up the pitch, which would inflate certain metrics. On the other hand, teams who are winning may not sit a lot deeper and concede more possession, therefore deflating their attacking metrics but inflate their defensive stats as they are inviting on pressure. For example, as Ivory Coast lost the first two games, it is possible that some of the metrics were impacted by the fact that they were chasing the game, which would have inflated their stats. Whereas with Faé, in the game against Mali they had a player who was sent off, meaning they were under a lot of pressure from the opposing team. Therefore, the defensive stats would be increasingly inflated because of this.

4 Methodology

4.1 Going from raw data to dashboard

Once the binary variables were created, it allowed for all the metrics, visualisations and tables to be split between each managerial era consistently. All the data was separated into their own figures, graphs and tables and then saved individually as RDS files using 'saveRDS()'. This meant that the dashboard was able to load the data using 'readRDS()'. This meant that instead of the dashboard having to process the raw data every time, it was able to simply read the pre-saved outputs. This was an important decision as it kept the dashboard running smoothly and avoided any issues in loading the data.

The visualisations were built using "ggplot2" first before being made interactive. This was done using 'ggplotly', which allowed for the shot maps to have more analytical information attached to them (player name, xG, minute and outcome). The passing and dribbling heat maps were kept as static 'renderPlot()' as they only showed aggregate density patterns which did not need to be interactive. There were issues with 'ggplotly' with 'final_shotmap.rds', which meant that a change back to a static plot was needed. They were then processed into Shinylive, where the dashboard was created. Once this was done, it was able to be deployed publicly on GitHub Pages.

4.2 Metrics computed

The analysis generated a wide range of metrics across attacking, progressive and defensive dimensions. All the metrics used directly contributed to the research question and directly speak to the change in performance between the two managers. The key metrics that were used were:

- xG per shot/game by manager - This is one of the most important metrics in this analysis as it directly answers whether Ivory Coast were creating better quality chances under Faé or Gasset, which was the main question being assessed. It captures the quality of the chances that have been created rather than whether they were scored. This is a more reliable indicator of attacking performance as goals scored can be heavily influenced by finishing inconsistencies, goalkeeper performances and moments of individual brilliance or errors.

- xG Difference per game by manager - This combined the attacking and defensive xG for both managers, and allowed us to compare how they performed against each other. Rather than estimating xG for and xG against only by themselves, the difference between the two condenses each managers overall performance into one singular figure. This made it useful to analyse the overall performance, seeing whether there was an actual improvement.

- Average Shots taken and conceded per game by manager - These were two more important metrics that enabled a clear analysis on the performances of the managers. Tying this up with the average xG generated from these shots allowed for efficient analysis on the quality of the chances, and whether or not they were limiting their opponents to dangerous or low quality chances.

- Win percentage per manager - This was used to establish an overview of the results before introducing the more complex metrics. Without knowing what the percentages of each manager were, the analysis would have very little context as numbers alone have very little meaning if those interpreting can't see how they related to on pitch results.
- Cumulative xG per round - This was used as per game figures do not show if there has been a fluctuation per game throughout the tournament, and it has the potential to hide anomalies (very high or very low xG matches).
- Goals and Assists per player - Used in the key player section as it is the most fundamental measure of direct attacking contribution before moving into the more complex metrics.
- Deserved winner score: attacking, progressive and defensive components - This was used because there are no singular metrics that can decide whether a team deserves to win a tournament. A composite score was calculated using all three components and a weighting system (decided by where they finished), to decide who deserved to win. The stage weighting was applied to take into consideration that the teams who progressed further into the tournaments are more likely to play tougher opponents.

4.3 Grouping

All the manager comparison metrics were computed on a game to game basis rather than a total number. This was because comparing the raw totals would have been misleading as Faé managed 4 games whereas Gasset managed 3. This would naturally mean that the manager with more games would accumulate a larger number, so per game rates was the best option for the analysis as it ensured it was not distorted and was fair.

For the deserved winner score, every per game metric was divided by the highest value that was recorded across the entire tournament before they were combined in the three scoring components. This was necessary because otherwise the scoring may have been distorted by very high values which may have increased the influence on the final result. Once this was done, a weighting system was applied depending on where each team had finished. The teams that went further into the tournament were given a higher weighting as the further you progress, the tougher the opponents get.

4.4 Dashboard Design

The dashboard was built in R Shiny using the 'bslib' R package to apply the themes and the custom background colour. The home page displays the research question at the top and has the four large navigation buttons which contain each piece of analysis. The four navigation buttons were 'Tournament Journey', 'Before and

After Manager Sacking', 'Key Players' and 'Was It Deserved?'. Each had its own colour which makes it easier to distinguish between the sections.

The navigation between each of the panels is handled using 'conditionalPanel()'blocks which show or hide each of the pages depending on which has been selected. 'ReactiveVal' stores which page is active, and a Javascript message handler updates it whenever something has been selected. This means that the pages are able to change smoothly without having to reload. Each page has it's own back button which allows for the user to go back to the home screen.

Within each of the pages, the analysis is split into different tabs using 'navset_pill()' to avoid unrelated information being placed in the same tabs. This meant that the analysis was in clean locations, and it would allow the user to find the exact information they need.

Within the dashboard, 'renderPlotly()'and 'renderDT()' were used so that the reactive graphs and tables were able to be interactive for the user. The shot map which compared the team's shots between each of the manager needed to be interactive so that the analysis would be evident within the map. For the table, it was made interactive so that the user can filter between each metric and see how was leading in each stat.

5 Findings

5.1 Tournament Overview and the Group Stage Struggle

Across the whole tournament, Ivory Coast played 7 matches, winning 4, drawing one (won the game via a penalty shootout) and losing twice. Their two defeats came in the group stage where they lost to Nigeria 1-0 and Equatorial Guinea 4-0 in the final group stage game, which triggered the managerial change. During the knockout stages, Ivory Coast won three games and drew one, which they won on penalties. They had performed at a much higher standard, and that will be analysed later in this section. Table 2 summarises Ivory Coast's full tournament timeline, showing who they had faced, what the score was and what the result of the game was.

Table 2: Côte d'Ivoire's match by match results across AFCON

competition_stage.name	opponent	ivc_score	opp_score	result
Final	Nigeria	2	1	Win
Semi-finals	Congo DR	1	0	Win
Round of 16	Senegal	1	1	Draw
Quarter-finals	Mali	2	1	Win
Group Stage	Equatorial Guinea	0	4	Loss
Group Stage	Nigeria	0	1	Loss
Group Stage	Guinea-Bissau	2	0	Win

The underlying data from these matches tell us a different story. As shown in @fig-cumulative-xG-chart, Ivory Coast had a steady accumulation of xG in the group stage. Across the entire group stage, they accumulated 3.39 xG per game, which comes to an average of 1.13 per game. Taking the results into consideration, we can see that they produced inconsistent performances throughout the group stage, with the two defeats. Despite losing 4-0 in the last game to Equatorial Guinea, the team had generated 2.06 xG, which is more than the first two games combined. The lack of xG within those first two games indicate a lack of attacking creativity and a difficulty in producing high quality opportunities. The majority of shots that were created accumulated a lower amount of xG, therefore indicating the likelihood of scoring was low. However in the last game of the group stage, despite the result, they managed to create the chances but were not able to finish them off. This sharp defeat led to the managerial change, even though they narrowly progressed to the next stage.

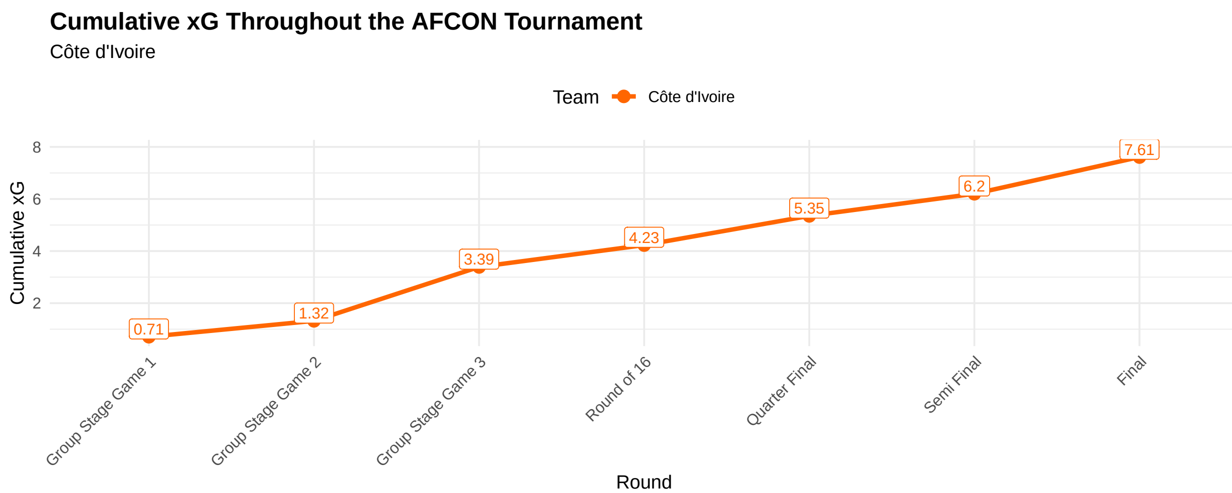

Figure 1: A graph highlighting the cumulative xG of Ivory Coast through each stage of the tournament

5.2 The Managerial Change: The introduction of Emerse Faé

The appointment of Emerse Faé was a surprise for many reasons. No one expected a team to dismiss their manager during the season, especially as they had reached the knockout stages. However, the performances from the team under head coach Jean-Louis Gasset caused for immediate change. The managerial change produced measurable data evidenced changes across all metrics of Ivory Coast's performance. An examination of six areas of comparison between the two managers will best highlight the significant impact that the new manager created.

5.2.1 Shot Map and Attacking Locations

The shot maps in Figure 2 offer the clearest immediate illustration of how the team's attacking intent had changed.

Under Emerse Faé, Ivory Coast's shots appear to be more centrally concentrated inside and around the penalty area. a large proportion of these shots came in a dangerous area, which therefore gave them a higher chance of scoring. This suggests that there was a more direct attacking structure, which was focused on creating higher quality opportunities in central areas for their attackers. Additionally, the concentration of attacks near the goal indicates that they were frequently able to progress possession in advanced areas, whilst being able to finish the attacks (scoring or missing the chance).

In contrast, under Jean-Louis Gasset the shot distance is noticeably wider and more dispersed across the box. There are more attempts from wider angles, which the StatsBomb xG model assigns that the probability of those chances to be below 0.1. Goals were rare for the team under Gasset, and the two goals that they scored came from chances accumulating 0.075 xg and 0.056 xg respectively.

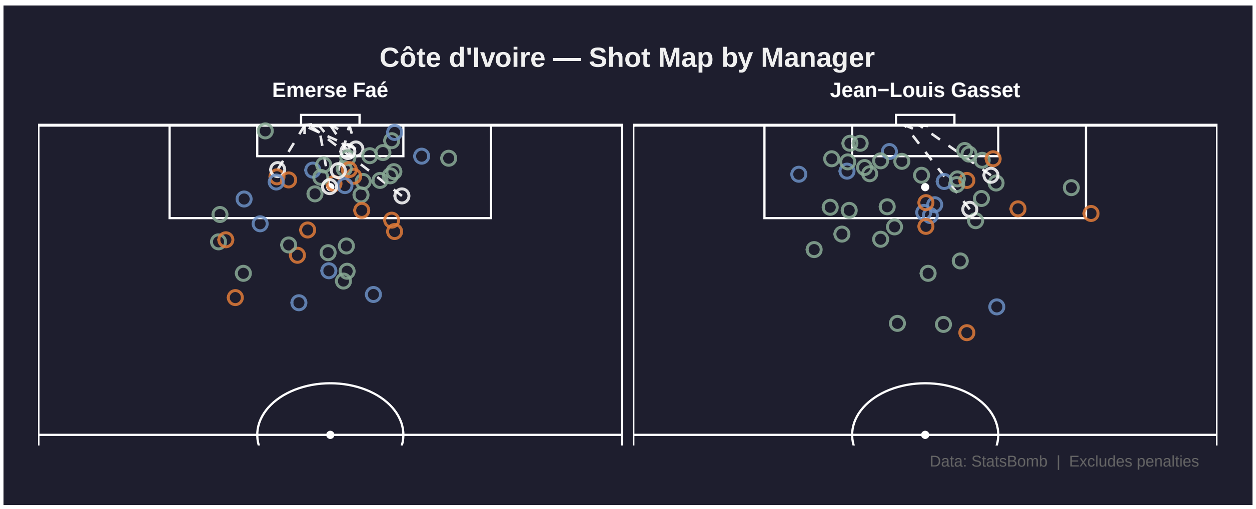

Figure 2: Comparison Between The Two Managers' Shot Maps

5.2.2 Shot Quality vs Quantity

The shot quality data shown in Table 3 formalises what the map shows in data form, illustrating a clear contrast between the two managers. Under Gasset, Ivory Coast averaged 15.3 shots per game at an average of 0.074 xG per shot. They had a shot on target percentage of 19.6%, with only 9 of their 46 shots being on target. Even though Jean-Louis Gasset's team averaged more shots per game, their output was far less effective. Faé's team scored 6 goals from their 51 shots, whereas Gasset's team only scored 2 from 46. These low amount of goals may not seem a lot, however, tournament football does not always provide high scoring games when teams of similar level play not to lose rather than win. There is a clear difference between avg xG per shot, where Faé's team averaged 0.083 per shot compared to Gasset's 0.074. On the surface level the difference may not seem that large, however over a large amount of shots it reflects a meaningful improvement in shot quality. This further supports the shot map illustrating that the shots under Faé were in more dangerous areas. '

Table 3: Shot Quality vs Quantity Comparison by Manager

Metric	Emerse Faé	Jean-Louis Gasset
Matches	4	3
Total Shots	51	46
Shots on Target	16	9
Total xG	4.22	3.39
Goals	6	2
Shots per Game	12.8	15.3
Avg xG per Shot	0.083	0.074
Shot Accuracy (%)	31.4	19.6

The difference in shot on target percentage is huge, as Faé's team has an average of 31.4% and Gasset's has an average of 19.6%. This means that nearly one third of shots by Faé's team tested the goalkeeper, while fewer than one in five did so under Gasset. An interesting contrast emerges between shot quantity and quality as Gasset's team attempted more shots but produced a lower shot on target percentage, lower average xG per shot, fewer goals and a lower total xG. This suggests that a higher shooting volume did not translate into higher attacking performance.

Overall, the table suggests that Ivory Coast were a more clinical and structurally efficient attacking side under Faé. Although Gasset's team had generated more shots in less games, they were lower in xG, suggesting the shots were taken with more pressure and in positions that are less likely to test the goalkeeper. In contrast, Fae's team created higher quality chances, tested the goalkeeper more frequently and converted their chances more efficiently.

5.2.3 Shots Conceded and xG Against and Defensive Actions

The data suggests that there was a clear improvement in defensive performance following the managerial change. Under Gasset, Ivory Coast conceded more shots per game, and xG per shot with 10.3 shots per game and 1.07 xG per shot. This indicates that they had a vulnerable defensive structure due to the higher amount of high quality chances that they conceded.

In contrast, under Faé, there was a slight improvement. Even though they conceded more xG and more shots, the average for both were lower as their shots conceded per game was 9 and 0.93 xG per shot. This illustrates that they restructured their opponents to lower quality chances, reflecting a more organised and resilient structure.

Table 4: Ivory Coast Shots Conceded and xG Against Table

Stat	Jean-Louis Gasset	Emerse Faé
matches	3	4

Stat	Jean-Louis Gasset	Emerse Faé
shots_conceded	31	36
xg_conceded	3.2	3.73
shots_conceded_per_game	10.3	9
xg_conceded_per_game	1.07	0.93

Furthermore, the defensive metrics used have reinforced the improvement in their structure. There has been a rise in tackles from 14.3 to 19.5, suggesting that they have gained more success in tackles and are defending more as a unit. This can be linked to pressures which has also increased from 104 to 141.8 per game. This indicates that there has been a change in intensity, with a more aggressive approach which allows them to turnover possession high, and start a counter attack using their quick wingers. Duels per game have also increased from 26.7 to 36.5, which reflects a more physical approach in contests and ball winning situations in midfield and defense. Interceptions have increased from 6.3 to 7.8 per game which indicates that there has been an increased focus on closing down spaces and blocking passing lanes. Finally, clearances have increased from 11 to 26 per game which suggests that the defenders are prioritising clearing the ball away from dangerous situations rather than keep it in the attacking zones.

```
readRDS("app/defensive_actions_manager_comparison.rds") |> knitr::kable()
```

Table 5: Defensive Actions between Managers

Stat	Jean-Louis Gasset	Emerse Faé
matches	3	4
tackles_pg	14.3	19.5
pressures_pg	104	141.8
duels_pg	26.7	36.5
interceptions_pg	6.3	7.8
clearances_pg	11	26

5.2.4 Passing Structure

?@fig-passing_heatmap compares heat maps of passes under each manager.

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i Please use `after_stat(level)` instead.
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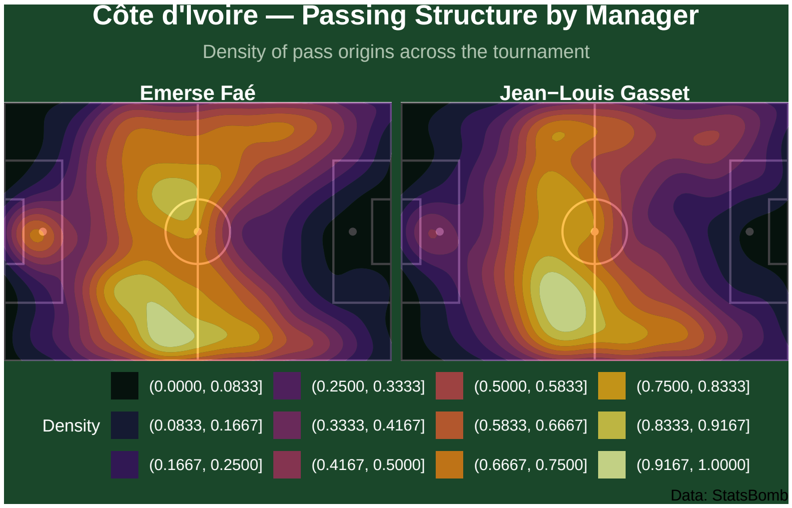

Figure 3: Passing Heatmap Comparison By Manager

Both heat maps are very similar with very high concentration in deeper and wider areas. This is further supported by the Table 8 which illustrates the full backs were very influential in progressive passes. The central concentration is higher under Faé, suggesting that the midfielders were receiving the ball in more advanced areas, suggesting a more attacking approach.

5.2.5 Dribble Areas

Figure 4 compares Ivory Coast's dribble density under each manager.

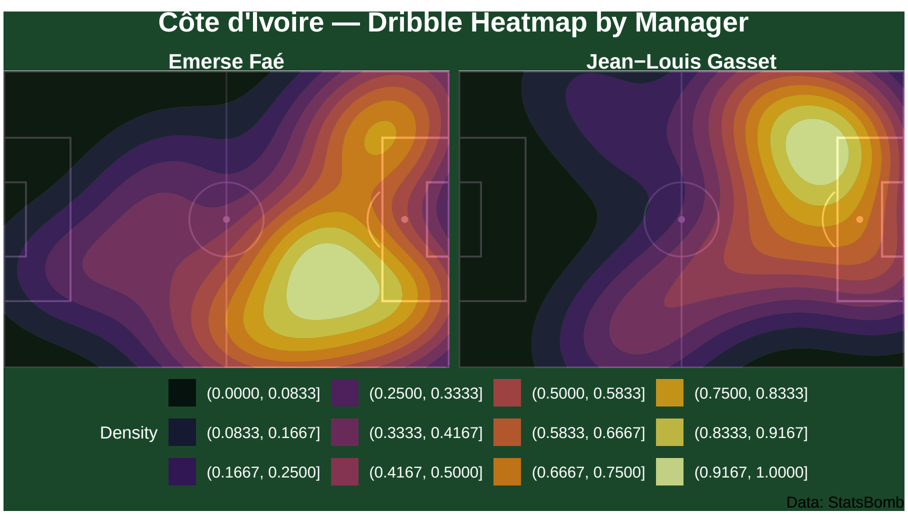

Figure 4: Dribble Heatmap Comparison By Manager

Under Faé, the dribble areas are heavily concentrated on the wide areas, with it being spread on both sides of the pitch. It is also very concentrated centrally just outside the box. This suggests that the majority of the dribbling was used as a way to break into dangerous areas, especially targeting the penalty area and isolating the fullbacks on both wings.

Under Gasset, there are very similar dribbling patterns where the wing areas have been targeted. However the major difference is the overreliance on one side. With Faé, there is high concentration on both sides, whereas under Gasset, they were only mainly targeting one side of the pitch.

5.3 Key player contributions

5.3.1 Shot Map

Figure 5 below illustrates the shot maps of the top five shooters for Ivory Coast.

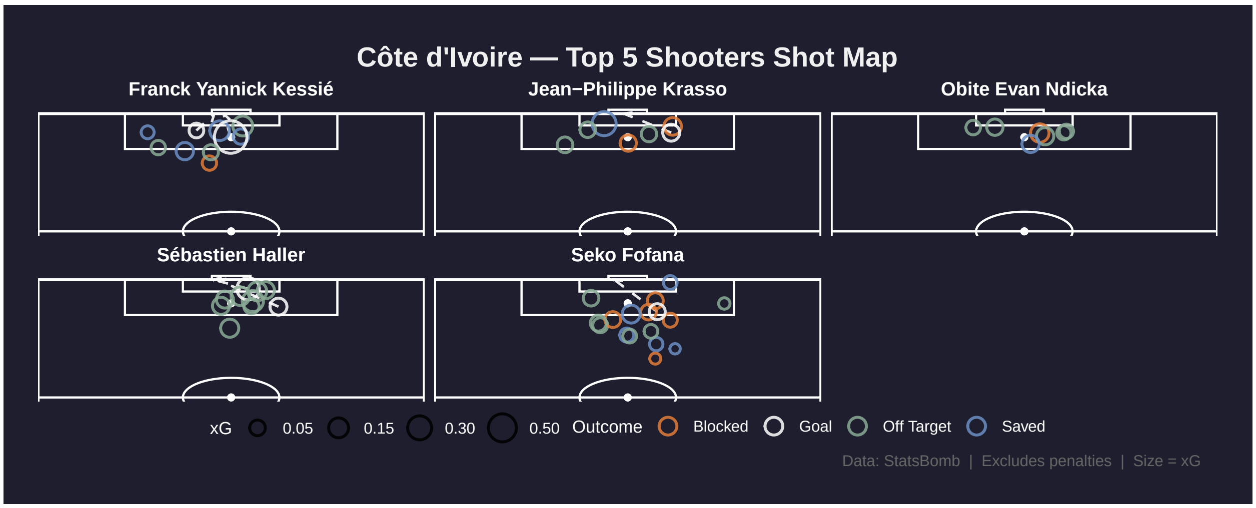

Figure 5: Top 5 Shooters' Heatmap for Ivory Coast

As shown, Sébastien Haller's shots are heavily concentrated very close to the goals, which highlights his role as the main striker and the main focal point for all the attacks. In contrast, we can see that the midfielders Franck Kessie and Seko Fofana show a much wider distribution of shots, as they are attempting shots from outside the box whilst also taking shots within the box. This indicates that they are making late runs into the box. Krasso also has shots in central areas as he was used as a backup striker to Haller. Evan Ndicka's limited amount of shots likely came from set piece situations.

Overall, it appears from the shot maps that Ivory Coast had a well organised attacking strategy with distinct player responsibilities. A focus on effective opportunity generation and attacking via the middle of the pitch is indicated by the concentration of shots through dangerous central areas.

5.3.2 Goals and Assists

Table 6: Combined Goals and Assists Table

player.name	Goals	Assists	GA
Franck Yannick Kessié	2	1	3
Sébastien Haller	2	0	NA
Jean-Philippe Krasso	1	0	NA
Oumar Diakité	1	0	NA
Seko Fofana	1	0	NA
Simon Adingra	1	2	3
Max-Alain Gradel	0	1	NA

From this goals and assists table, we can see that the goals have been spread across the team, as no player has scored more than two goals. Franck Kessié and Simon Adingra recorded the highest amount of goal contributions with three each, which highlights their importance in both chance creation and goal scoring. Overall, the table shows that the team was very balanced in attack, and they did not have to rely on one singular player.

5.3.3 Dribbling Heat Maps

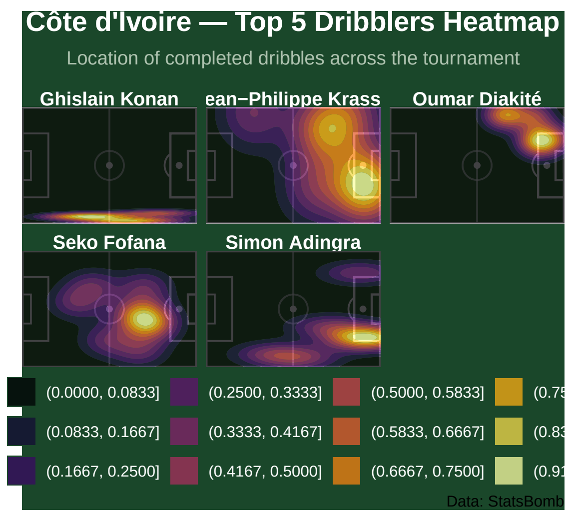

Figure 6: Heatmap of Top 5 dribblers for Ivory Coast

The dribbling heat maps show that Ivory Coast relied heavily on wide attacking dribbles. Oumar Diakité, Jean-Philippe Krasso and Simon Adingra are the three players with the highest concentration in the wider areas. This further supports the point that they relied on wing play to attack the opponent and pin them back.

In contrast, we have Seko Fofana who plays in midfield meaning his dribbling concentration is more central. He used his dribbles to connect the defense and the attack, and progress the ball through midfield. Ghislain

Konan showed deeper and wider dribbling on the left side of defense, which illustrated his importance in helping the team relieve pressure by dribbling out of tough situations.

5.3.4 Defensive stats

The defensive data highlights that Kessié and Seri were the most efficient defensive midfielders. Kessié recorded a very strong tackle success rate at 83.3% whilst Seri recorded 80%. These were both from a high number of tackles meaning that they were very efficient in the defensive roles that had been set for them. Sangaré also contributed massively, not only with tackles. All round he was a very solid defensive midfielder which allowed the attackers freedom to play.

In the wider defensive areas, both Konan and Singo were involved in a large number of duels and recoveries. Both recorded rather low percentages for their duels won, however, they both excelled at other areas such as interceptions and clearances. The low percentage duels won indicate they were facing difficult one on one situations.

Overall, the table highlights that their defensive strength came from the effectiveness of their midfield, which was supported by the aggressive nature of their fullbacks.

Table 7: Ivory Coast Defensive stats by players

player.name	tackles	tackles_won	tackle_pct	pressures	interceptions	inter_success	inter_pct	duels	duels_won	duel_pct	clearances
Ghislain Konan	19	8	42.1	66	8	5	62.5	25	8	32.0	15
Franck Yannick Kessié	12	10	83.3	89	4	3	75.0	17	10	58.8	7
Ibrahim Sangaré	10	7	70.0	96	4	1	25.0	13	7	53.8	10
Jean Michaël Seri	10	8	80.0	51	3	3	100.0	11	8	72.7	3
Wilfried Stephane Singo	9	3	33.3	36	7	5	71.4	15	3	20.0	18

5.3.5 Top 5 Passers

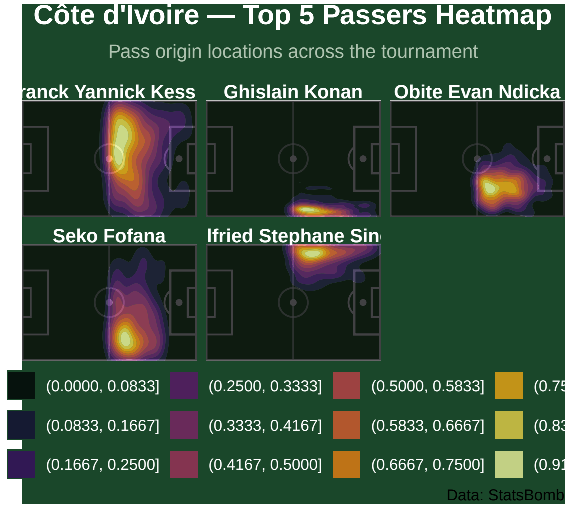

Figure 7: Heatmap of Passing Areas for Ivory Coast

The passing heat maps indicate that Ivory Coast's buildup play was heavily concentrated through central areas. Kessié and Fofana illustrated high passing activity in central areas, which highlights their importance in connecting the attack and defense with their progressive passes.

Distribution from the flanks was also important for Ivory Coast with Singo and Konan both having high concentration in wide areas. The teams use of fullbacks to support buildup allowed them to maintain width in attacks. Furthermore, Evan Ndicka had a huge role in progression as the high concentration suggests he had a key role in starting attacks.

5.3.6 Progressive Stats

This attacking data table highlights that Seko Fofana is one of the most influential offensive players. He ranks highly in four out of the 6 metrics, with only progressive passing and box entries being his weak points. Simon Adingra and Oumar Diakaté both played major roles in attacks, with high amounts in all metrics apart from progressive passes which indicates that they are always on the receiving end of progressive passes.

There is a huge reliance on the fullbacks in buildup with Konan, Aurier and Singo who have high entries into and touches in the final 3rd, with a huge amount of progressive passes. This further indicates that there was a huge reliance on width, and the players involved supported efficiently.

Table 8: Ivory Coast Progressive Metrics

player.name	carries_final3rd	box_entries	dribbles	touches_final3rd	progressive_passes	shot_assists
Ghislain Konan	12	0	5	163	150	5

Table 8: Ivory Coast Progressive Metrics

player.name	carries_final3rd	box_entries	dribbles	touches_final3rd	progressive_passes	shot_assists
Seko Fofana	15	1	9	160	49	5
Oumar Diakité	7	3	6	146	12	2
Franck Yannick Kessié	10	1	4	135	75	6
Serge Aurier	6	0	1	133	74	1

5.4 Was the title deserved?

A metric was created to determine who deserved to win the tournament. A weighting was applied depending on what stage the team reached, with the final having a weight of 1, semi final 0.9, Quarter Final 0.85, Round of 16 0.8 and the Group Stage 0.75. This was done so that all factors could be taken into consideration, such as the amount of games they had to play and teams playing harder opponents later in the tournament. Attacking, progressive and defensive stats were also included in the metric.

Table 9: Tournament's Deserved Winner

team.name	competition_stage	stage_weight	games	goals	xg_pg	shots_on_target_pg	shot_assists_pg	progressive_passes_pg	dribbles_pg	carries_final3rd_pg	box_entries_pg	touches_final3rd_pg	tackles_pg	interceptions_pg	clearances_pg	pressures_pg	attack_score	progressive_score	defensive_score	overall_score
Côte d'Ivoire	Final	1.00	7	8	1.09	3.6	9.4	165.9	8.0	13.9	3.6	240.0	10.1	7.1	19.6	125.6	6.3	8.0	6.1	6.8
Egypt	Round of 16	0.80	4	7	1.57	5.2	12.2	163.2	9.0	18.5	4.8	277.0	9.0	10.8	19.8	143.8	7.1	7.4	5.6	6.7
Nigeria	Final	1.00	7	7	1.39	3.1	7.6	138.1	6.3	11.4	4.7	184.4	9.3	7.4	20.7	141.3	5.9	6.9	6.4	6.4
Cape Verde Islands	Quarterfinals	0.85	5	8	1.36	3.6	8.6	167.2	9.2	9.2	4.2	201.4	11.4	9.6	24.0	144.0	5.9	6.5	6.3	6.2

Table 9: Tournament's Deserved Winner

team.name	competition_stage	stage_weight	matches	goals	xg_pg	shots_on_target_pg	shot_assists_pg	progressive_passes_pg	dribbles_pg	carries_final3rd_pg	box_entries_pg	touches_final3rd_pg	tackles_won_pg	interceptions_pg	clearances_pg	pressures_pg	attack_score	progressive_score	defensive_score	overall_score
South Africa	Semi-final → 3rd Place Final	0.90	7	7	1.07	4.6	6.9	175.7	7.3	15.7	3.6	203.4	7.7	8.7	22.6	149.3	5.5	7.2	6.0	6.2

The scores revealed that Ivory Coast did in fact deserve to win the tournament despite their poor Group Stage performance as their underlying metrics improved significantly across the tournament. Their transition from the inconsistent group of performers to a more efficient team in the knockout stages is reflected in the high attacking score. All round, they are the most consistent team across the four key attributes.

Nigeria who also got to the final are in third place, suggesting that Ivory Coast's win in the final was deserved. While Nigeria were more consistent throughout the tournament, Ivory Coast's effective knockout stage, tied with the weighting of reaching the final, allowed for them to finish at the top of this chart.

Interestingly, Egypt ranked second in this chart despite finishing in the Round of 16. There are many factors to this as they were very strong in the group stages, however, key player Mohammed Salah suffered an injury meaning they were missing their strongest player for the knockout stages.

6 Discussion, Limitation and Conclusion

6.1 Implication of the Findings

The findings reveal that the managerial change from Gasset to Faé had a measurable impact on Ivory Coast's underlying performance metrics rather than just the results. There was an improvement in their shot quality as their xG per shot had risen from 0.074 to 0.083. Even though it is a small increase, over an increased number of shots, the margin becomes bigger and bigger. This combined with the increased shot on target percentage rose from 19.6% to 31.4%, suggesting that the team were not only creating better opportunities, they were also testing the goalkeeper a lot more. The defensive data also illustrates the improvement as the increase suggests that there was a structural change which solidified their team. The improvement in attacking and defensive metrics were key to them winning the tournament, and the managerial change helped manage that.

The findings from this analysis have different implications for different audiences.

Firstly for coaches and sporting directors, the data demonstrates that a change of manager during a tournament can produce improvements on the pitch and in underlying performances metrics. The improvement seen under Faé practically won Ivory Coast the tournament as they were on the brink of elimination under Gasset. These improvements were visible with the results and multiple metrics, such as the shot locations became more central which led to an improvement in their shot quality. The dribbling and passing structures had shifted into more advanced areas of the pitch which indicates that they had a more attacking structure and were creating more dangerous attacks. This on the field improvement could impact future decisions about whether making a managerial change during a competition will benefit or hinder the team involved.

For analysts, the dashboard offers a way of looking at Ivory Coast's tournament performance beyond just their results and basic stats. The manager comparison section shows how event level data can be used to build a more complete analysis of how a team has performed. The data shows that Ivory Coast had an improvement after the managerial change, which in the end enabled them to win the competition. The shot metrics, defensive actions tables and the spatial heat maps provide the analysts a level of detail which results based analysis cannot offer. This would allow analysts to pinpoint exactly where they have improved and what they can improve on. This kind of approach could easily be applied to other teams or competitions where the event level data is available for use. Whether it is assessing the impact of a tactical change during the season, analysing if a strong run of results is genuinely supported by the underlying performance data, or identifying which players are most important to a team's overall structure and style of play, the same methodology used in this project could easily be adapted to explore those questions in a similarly effective way.

6.2 Limitations

One limitation of this analysis is that the StatsBomb data does not include tracking data for off the ball movement. This would be particularly relevant for the defensive analysis when comparing the two managers as these metrics such as pressing intensity and defensive line height cannot be measured from event level data alone. The improvement between the two managers can only be calculated through on-ball actions like tackles, pressures, interceptions and clearances rather than a direct observation of how the defense as a whole were structured.

Another limitation of the analysis is the deserved winner table. Even though the deserved winner score placed Ivory Coast at the tournament rankings, the weightings assigned and the metrics used were all judgement calls that others may use differently. The choices that have been made are reasonable and bode well with the data that was available, however, if other data became available, then the metrics and the weighting could be rebuilt. In regard to the weighting, it was based on how far a team reached in the tournament, due to the opposition being more difficult later. Being able to compute the difficulty of each opponent and using that average as the weighting could be something used with different data.

6.3 Future Outlook

There are several ways this analysis could be improved for future outlook. Firstly, with the access to additional data, StatsBomb's freeze frame data which records the positions of all the 22 players at the moment of each event would have allowed for a much more precise analysis (StatsBomb, 2024). This would have enabled an analysis of the defensive shape and the quality of the shots in relation to where the defenders are positioned. This gives a clear picture of how dangerous the chance is, as shots where the attacker is surrounded by no one and one where they are under pressure are completely different. This is something that normal xG models cannot capture.

6.4 Conclusion

This report set out to answer a single question: how did Ivory Coast win AFCON 2023 despite sacking their manager during the tournament, and does the data back them up?

The data has provided a clear answer to this question. The managerial change from Jean-Louis Gasset to Emerse Faé produced a measurable improvement across all metrics of performance that have been examined. Average xG per shot rose from 0.074 to 0.083, shot percentage almost doubled from 19.6% to 31.4%, tackles had increased from 14.3 to 19.5 per game and pressures had increased from 104 to 141.8 per game. Alongside this, shots conceded and xG conceded per game also decreased. As a squad, their overall goal contributions were spread across their team, with no standout performer. When combining all the metrics, including the weighting score, in the deserved winner table, Ivory Coast were placed first in the ranking, whilst other finalists Nigeria placed in third. The conclusions from this table should be held with caution due to the weighting score used were all judgement calls and not the most complete way of weighting. This is due to not having enough data to judge the difficulty of the opponents, where the weighting used in this analysis was on how far a team went in the tournament.

The game state limitation is also worth noting, as Ivory Coast would have been chasing the games as they were losing, meaning the data from the matches may have been inflated in attacking metrics and deflated in defensive metrics. This means that the gap between the two managers may not be quite as large as the raw data may suggest.

Nevertheless, the raw data points to the conclusion that Ivory Coast won because of the managerial change. Faé had transformed a team that were struggling to create many high quality chances into one that was more direct and more clinical in front of goal. Defensively they became more organised and a much harder team to beat. Overall, the data illustrates that they deserved to win the title, and the change in manager was the biggest cause of this.

7 List of AI tool used

- Claude (Anthropic, 2025) . Claude.

  - Used to help clean up and develop code. Helped with errors and refining code used

  - Used to proofread to check my mistakes in writing

- Grammarly Inc. (2025). Grammarly

  https://www.grammarly.com/

  - Support with grammar and structure of academic writing

- ChatGPT: Open AI (2025)

https://chat.openai.com

- Assisted in paraphrasing, creating synonyms and helping structure certain paragraphs.

8 References

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